

Time & Work

12 july,2022 I shift

1. 24 Woman and 9 men can complete a certain piece of work in 18 days. Each woman takes twice the amount of time taken by a man to finish the work. In how many days can only 9 men finish the same work?

Solution:

Let each man take x days to complete the work, then a woman will take $2x$ days to complete the work.

According to the question,

$$\Rightarrow \frac{9}{x} + \frac{24}{2x} = \frac{1}{18}$$

$$\Rightarrow \frac{18+24}{2x} = \frac{1}{18}$$

$$\Rightarrow \frac{42}{2x} = \frac{1}{18}$$

$$\Rightarrow X = 42 \times 9 = 378$$

$$\Rightarrow \text{Now, 9 men will complete the work in } \frac{378}{9} = 42 \text{ days}$$

2. Bobby and Billy can complete a job in 28 days and 20 days, respectively. They work together for 5 days, after which Billy leaves the job. In how many days did Bobby alone complete the rest of the job?

Solution:

$$\Rightarrow \text{Bobby and Billy's one day work} = \left(\frac{1}{28} + \frac{1}{20} \right)$$

$$\Rightarrow \text{Bobby and Billy's 5 day's work} = 5 \left(\frac{1}{28} + \frac{1}{20} \right) = \frac{5(5+7)}{140} = \frac{3}{7}$$

$$\Rightarrow \text{Remaining work} = 1 - \frac{3}{7} = \frac{4}{7}$$

Let Bobby alone complete the remaining work in x days.

$$\text{So, } x \times \frac{1}{28} = \frac{4}{7}$$

$$** X = \frac{4}{7} \times 28 = 16 \text{ days}$$

3. Zee and Gee can complete a task together in 24 days. If the efficiencies of Zee and Gee are in the ratio 3:8 respectively, then in how many days can Gee complete the task alone.

Solution:

According to the question

$$\begin{aligned}\text{Total time taken by Gee to complete the task alone} &= \frac{24(3+8)}{8} = 3 \times 8 \\ &= 33 \text{ days}\end{aligned}$$

4. D_1 and D_2 can do a piece of work together in 16 days, D_1 and D_3 can do the same work it together 24 days, while D_3 and D_1 can do it together in 48 days. In how many days can all three, working together, do $\frac{5}{6}$ of the work?

Solution:

According to the question

$$2 \times \left(\frac{1}{D_1} + \frac{1}{D_2} + \frac{1}{D_3} \right) = \left(\frac{1}{16} + \frac{1}{24} + \frac{1}{48} \right) = \frac{3+2+1}{48} = \frac{6}{48}$$

$$\Rightarrow \frac{1}{D_1} + \frac{1}{D_2} + \frac{1}{D_3} = \frac{6}{48} \times \frac{1}{2} = \frac{1}{16}$$

$$^{**} \text{ Required number of days} = 16 \times \frac{5}{6} = \frac{40}{3} \text{ days.}$$

5. A, B and C together can do a piece of work in 6 days. A and B together can do the same work in 12 days. B alone can do the same work in 18 days. In how many days can A and C together complete the same work?

Solution:

According to the question

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{6} \quad \text{.....(i)}$$

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{12} \quad \text{.....(ii)}$$

From equation (i) and (ii)

$$\frac{1}{C} = \frac{1}{6} - \frac{1}{12} = \frac{2-1}{12} = \frac{1}{12} \dots\dots\dots(\text{iii})$$

$$\text{And } \frac{1}{B} = \frac{1}{18} \dots\dots\dots(\text{iv})$$

From equation (iii) and (iv)

$$\frac{1}{A} = \frac{1}{36}$$

$$1 \text{ day's work of A and C} = \frac{1}{36} + \frac{1}{12} = \frac{1+3}{36} = \frac{1}{9}$$

**** A and C together can do the work in $\frac{1}{\frac{1}{9}} = 9$ days**

6. A can do a piece of work alone in 40 days. A and B together can do the same work in 10 days. C is 50% less efficient than B. In how many days can A and C together complete the same work?

Solution:

Given,

$$\frac{1}{A} = \frac{1}{40} \dots\dots\dots(\text{i})$$

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{10} \dots\dots\dots(\text{ii})$$

From equation (i) and (ii)

$$\frac{1}{B} = \frac{1}{10} - \frac{1}{40} = \frac{4-1}{40} = \frac{3}{40}$$

$$\text{Given } \frac{1}{C} = \frac{1}{A} = \frac{3}{80} \dots\dots\dots(\text{iii})$$

Now, from equation (i) and (iii)

$$\frac{1}{A} + \frac{1}{C} = \frac{1}{40} + \frac{3}{80} = \frac{2+3}{80} = \frac{5}{80}$$

**** Time taken by A and C together to complete the work**

$$= \frac{80}{5} = 16 \text{ days.}$$